

SASTRA DEEMED UNIVERSITY

(A University under section 3 of the UGC Act, 1956)

End Semester Examinations

July 2022

Course Code: **BIT101R01**

Course: **BIOLOGY FOR ENGINEERS**

Question Paper No. : **U0162**

Duration: **3 hours**

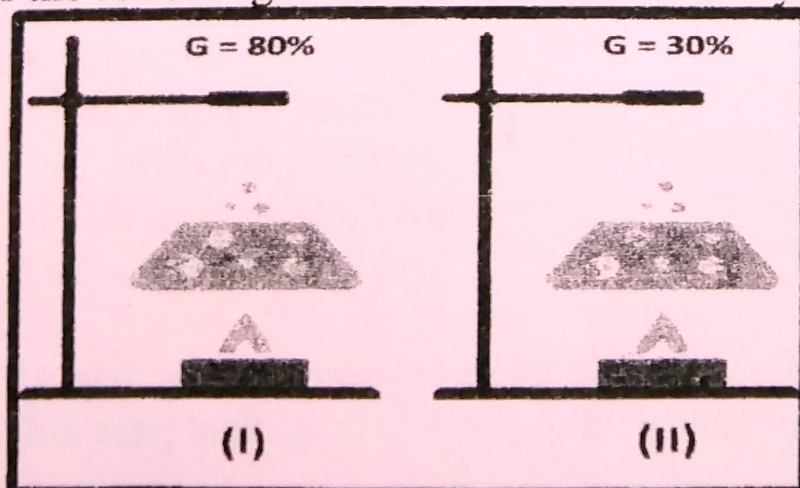
Max. Marks: **100**

PART – A

Answer all the questions

10 x 2 = 20 Marks

1. Life on earth has evolved for a very long time that caused the emergence of robust system. Therefore, biology has inspired engineers to innovatively design biomimetic products which are robust. Mention any two biomimetic products each inspired from ants and birds.
2. Given below is an illustration that depicts DNA containing solution that is heated. The DNA in the flask (I) consists of 80% guanine and the flask (II) contains 30% guanine nucleotides. The DNA molecules in which of the following flasks denature first? Justify your answer.



3. In the year 2018, twelve members of a football team in Thailand entered a cave for a birthday celebration. Unfortunately, they were soon blocked inside due to incessant rains and flooding. The rescue effort involved more than 10,000 people, including experts across the world. Assume you were given the task to rescue a similar incident with the help of robots. What kind of robots will you design and why?

4. Choose the correct options:
 - (a) Basic teaching and research with non-pathogens are carried out in BSL-1.
 - (b) Basic teaching and research with non-pathogens are carried out in BSL-3.
 - (c) Basic teaching and research with non-pathogens are carried out in BSL-4.
 - (d) Human/animal anatomical wastes are disposed in black colored bin.
 - (e) Human/animal anatomical wastes are disposed in yellow colored bin.
 - (f) Human/animal anatomical wastes are disposed in red colored bin.

5. A cell is essentially an industry that makes-up and breaks-up polymeric substances through dehydration and hydrolysis reactions. However, reactions take very long time to occur, ceasing the possibility of emergence of life on earth. For example, a study in 2008 showed that vital reactions in a cell would take 2.3 billion years to occur without enzymes. What then is the crucial thing that enzyme does to significantly accelerate a reaction?

6. Match the following:

(a) Pit viper	(i) rods to cone ratio = 1000:1
(b) bats	(ii) rods to cone ratio = 20:1
(c) Nocturnal animal	(iii) Echolocation
(d) Diurnal animal	(iv) infrared radiation
	(v) mechanoreceptor

7. A novel drug for a disease takes anywhere more than 15 years to be approved. Also, nearly 95 out of 100 drugs that pass animal tests fail

in humans. Suggest a cheaper alternative that is faster and more accurate than the animal studies.

8. A gene has this sequence 5'-ATG TTC CTC AGG GCC GCG TAG-3'. Which of the following tRNA anticodons will be chosen in the process of translation?
- (a) UCC
 - (b) AAG
 - (c) UUC
 - (d) UAG
 - (e) CUC
9. Most of us in the world are using smartphones and many are using computers nowadays. Therefore, the traditional method of storage cannot meet the requirement of big digital data storage. Which biomolecule will be your choice to meet this demand? Justify your answer.
10. Which of the following is true?
- (a) An artificial ventilator work on negative pressure.
 - (b) An artificial ventilator work on positive pressure.
 - (c) Human lungs work on positive pressure.
 - (d) Human lungs work on negative pressure..

PART – B

Answer any **FOUR** questions

4 x 15 = 60 Marks

11. (a) Assume you are hungry and went to the canteen to eat. A list of available food items is displayed in the menu card. You have decided to buy dosa. Explain the neuronal process that facilitated you to make this decision, particularly the structure of a neuron and the triggering of action potential. (10)
- (b) Explain the structure of ANN and how decision-making in it is implemented. (5)
12. Currently heart transplantation is done by harvesting the heart from a donor, who happens to be brain-dead. The harvested heart is

transported to long distances by creation of a green channel (traffic-free), indicating logistical challenges. According to estimates, there is a severe shortage in hearts and a long waitlist for getting the transplant. As a student of 'Biology for Engineers' course, suggest solutions to overcome the (a) logistical challenge and (b) shortage of the organ challenge.

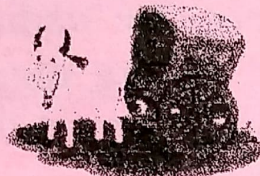
13. (a) Immunology principles have inspired engineers to design several algorithms for different applications. Describe the principles of adaptive immune system. (10)

(b) List at-least five different applications where AIS can be used. (5)

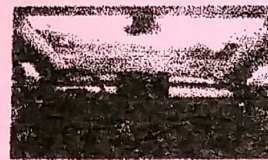
14. (a) Group these and connect them according to timeline: (3)



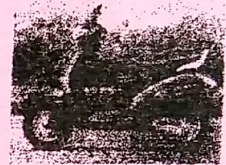
Bajaj scooter



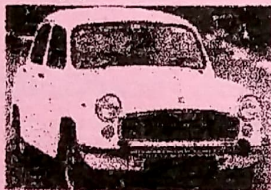
Bullock cart



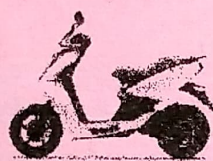
Self-driving car



Honda activa scooter



Ambassador



Ather electric scooter



Cycle



Horse cart

(b) In the above figure, some modes of transport have been endangered and is on the verge of extinction. Some are currently selected and some variants may possibly get selected in future. There are similarities of this to Darwinian evolution of life. Explain the defining principles of theory of natural selection. (6)

(c) Darwinian evolutionary theory has inspired the development of optimization programs like Genetic Algorithm (GA). Mention the structure of GA and particularly explain importance of the step that introduce variations. (6)

15. (a) A mutation in the DNA can cause protein dysfunction. For example, change in a single nucleotide causes sickle cell anemia. Why do you think DNA sequence fidelity is important for a protein function? (10)

- (b) Highest number of people globally die due to cardiac illness. One of the major reasons causing cardiac disease is the formation of plaques, which are formed by fatty substances, cholesterol, etc. We can substantially bring down the number of heart attacks, if an early detection of causative substances of vascular plaques can be done. Design a biosensor for early diagnosis of cholesterol in blood. (5)

16. (a) Carbohydrates are used by cells for (i) fuel, (ii) storage, and for (iii) maintaining its structural integrity. Discuss types of carbohydrates with examples for each type of function mentioned above. (9)
- (b) Describe the prokaryotic cell structure with a diagram. (6)

PART – C

Answer the following

1 x 20 = 20 Marks

17. Provide your bioinspired technological ideas for these challenges:
- (a) Milk has been an indispensable part of our food culture as it contains vital nutrients. However, adulteration of milk is a common problem. Can you suggest rapid detection methods of milk adulteration based on our olfactory and gustatory system?(4)
- (b) Many of our army personnel lose their limbs during war. Design a biomimetic technology for our personnel. (4)
- (c) For space exploration, designing a spacecraft with minimum payload is a challenge. Large-volume and heavy materials can limit the mission capabilities, whereas lightweight and compact technologies enable missions to spend more time in space. It is known that long-term vacuuming does not affect actuators that have ionic liquid electrolytes. Taking cue from the last sentence, can you suggest a bioinspired space-worthy technology for this task? (4)
- (d) It is estimated that more than 7 million people in India are with chronic kidney disease. Hemodialysis is the current solution to this problem. However, the patient needs to go nearly twice a week to a dialysis centre to filter the blood, which is not

economical and psychologically stressful. As a bioinspired engineer, can you suggest a solution to this problem? (4)

- (e) Sanitation workers are forced to enter toxic manholes to ensure that the drainage system works well in a city. Biorobots can potentially be manufactured for this task. Suggest prominent features that the robot must have to carry on this job? (4)
